DAI 1000
Digital and AI Literacy
L11 - AI Ethics and Social Impacts

Class Objectives

- **Ethics Guidelines for Trustworthy AI**
  - AI Privacy Issues
  - AI Fairness Issues
- Case Study

Ethics Guidelines for Trustworthy AI

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Ethics guidelines for trustworthy AI

The aim of the Guidelines is to promote Trustworthy AI. Trustworthy AI has three components, which should be met throughout the system's entire life cycle: (1) it should be lawful, complying with all applicable laws and regulations (2) it should be ethical, ensuring adherence to ethical principles and values and (3) it should be robust, both from a technical and social perspective since, even with good intentions, AI systems can cause unintentional harm. Each component in itself is necessary but not sufficient for the achievement of Trustworthy AI. Ideally, all three components work in harmony and overlap in their operation. If, in practice, tensions arise between these components, society should endeavour to align them.

View less

Seven Key Requirements

1 • **Human agency and oversight**

AI systems should **support human autonomy and decision-making**.

Users should be given the **knowledge and tools to comprehend and interact with AI systems** to a satisfactory degree and, where possible, be enabled to reasonably **self-assess or challenge the system**.

2 • **Technical robustness and safety**

Technical robustness requires that AI systems be developed with a **preventative approach to risks** and in a manner such that they reliably behave as intended while **minimizing unintentional and unexpected harm,** and preventing unacceptable harm.

In addition, the physical and mental integrity of humans should be ensured.

Seven Key Requirements

3 • **Privacy and data governance**

Prevention of harm to privacy necessitates **adequate data governance** that covers the integrity of the data used, its access protocols and the capability to process data in a manner that **protects privacy**.

4 • **Transparency**

The data sets and the processes that yield the **AI system's decision** (e.g., data gathering, data labelling and the algorithms used) should be **traceable**.

Moreover, it includes the **explainability** which concerns the ability **to explain both the technical processes of an AI system and the related human decisions**.

Seven Key Requirements

5 • **Diversity, non-discrimination and fairness**

It must enable **inclusion and diversity** throughout the entire AI system's life cycle.

Besides **the consideration and involvement of all affected stakeholders** throughout the process, this also entails **ensuring equal access through inclusive design processes as well as equal treatment**.

6 • **Societal and environmental well-being**

Sustainability and ecological responsibility of AI systems should be encouraged, and research should be fostered into AI solutions addressing **global concern (e.g., the Sustainable Development Goals)**.

Ideally, AI systems should be used to benefit all human beings, including future generations.

Seven Key Requirements

7 • **Accountability**

The requirement of accountability complements the above requirements, and is closely linked to the principle of fairness.

It necessitates that mechanisms be put in place to ensure **responsibility and accountability for AI systems and their outcomes**, both before and after their development, deployment and use.

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AI Privacy Issues: Sensitive Data Collection

The AI boom, including large language models and associated chatbots, poses unprecedented challenges for privacy.

Collection of Sensitive Data

AI poses greater data privacy risks than earlier technologies due to the **sheer volume of sensitive information** used as training data:

- Terabytes or petabytes of text, images, and video
- Healthcare information
- Personal data from social media
- Financial data
- Biometric data

AI Privacy Issues: Consent in Data Collection

Collection Without Consent

Controversy arises when data is procured for AI development without express consent or knowledge of the people from whom it's collected.

Example: LinkedIn faced backlash after users discovered they were automatically opted into allowing their data to train generative AI models.

Usage Beyond Initial Consent

Even when data is collected with consent, privacy risks emerge if used beyond initially disclosed purposes.

Example: In California, a former surgical patient discovered photos related to her medical treatment had been used in an AI training dataset without proper consent for that specific use.

AI Privacy Issues: Data exfiltration and Leakage

Sensitive Data Targets

AI models contain a trove of sensitive data that can prove **irresistible to attackers**.
Bad actors can conduct such data exfiltration (data theft) from AI applications through various strategies.

Example: In **prompt injection attacks**, hackers could disguise malicious inputs as legitimate prompts, manipulating generative AI systems into exposing sensitive data.

Accidental Data Leakage

Some AI models have proven **vulnerable to such data breaches**.
Consider a healthcare company that builds an AI-powered diagnostic app based on its customers' data. That app might unintentionally leak customers' private information to other customers who happen to use a particular prompt.

Example: In one headline-making instance, ChatGPT showed some users the titles of other users' conversation histories.

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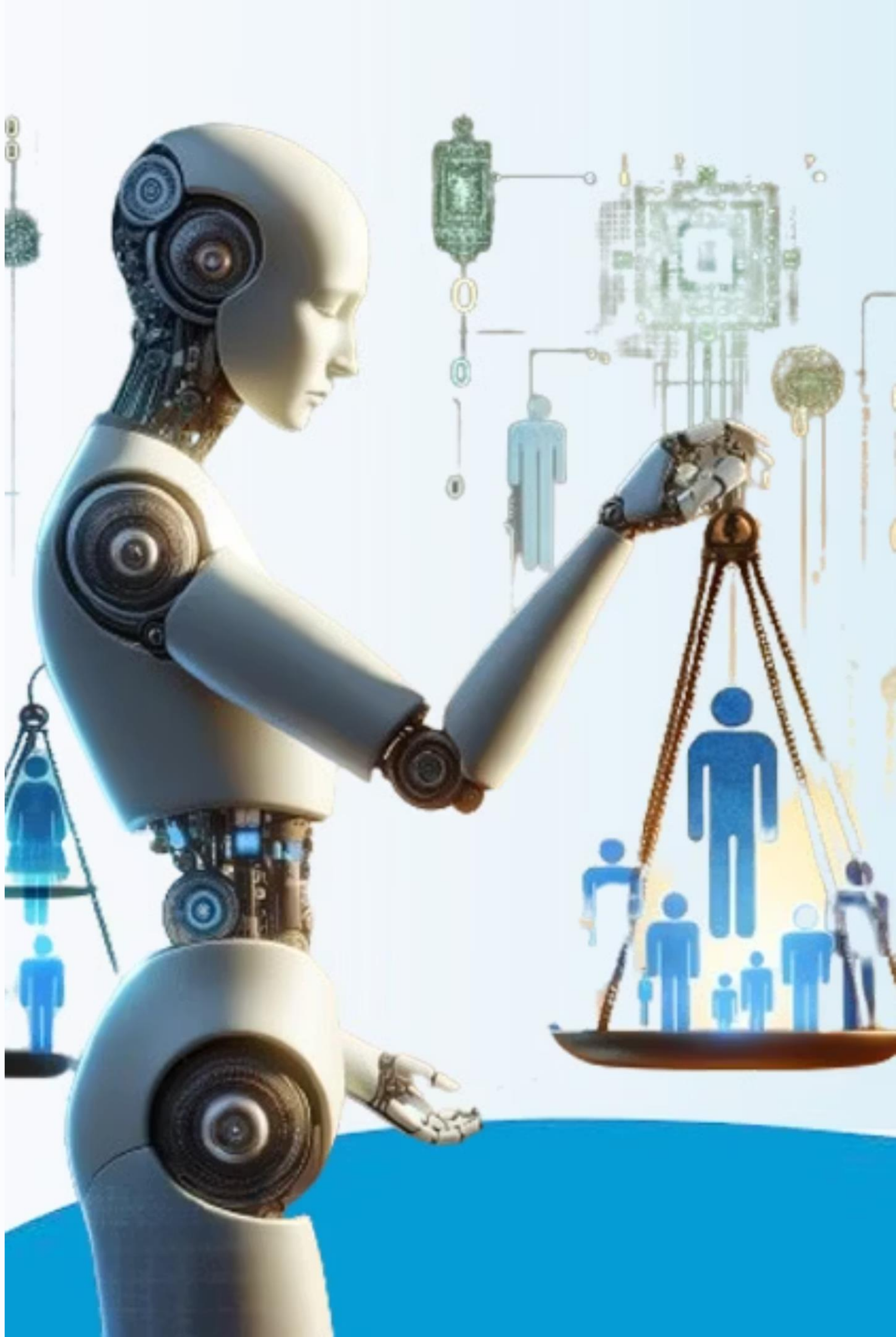
AI Fairness Issues

AI is augmenting and sometimes replacing human decision-making in important situations:

- who should get the job
- whose loan gets approved
- who gets pretrial release or parole …

However, machine learning models can perpetuate societal bias in these decision-making processes.



AI Fairness Examples

Unfair Criminal Justice Decisions

Predictive policing algorithms have disproportionately targeted communities of color, reinforcing systemic biases.

COMPAS Criminal Justice Algorithm

Used in the U.S. for assessing the risk of reoffending, the algorithm **disproportionately labeled Black defendants as high-risk compared to white defendants with similar records**.

AI Fairness Examples

Inequitable Loan Approvals

AI-driven lending systems have been criticized for **systematically rejecting loan applications from minority groups**.

In a study evaluating chatbot loan suggestions for mortgage applications, researchers found clear racial bias, with chatbots **recommending higher interest rates for Black applicants** and **labeling Black and Hispanic borrowers as "riskier."**

8.5%

Approval Gap

White applicants were 8.5% more likely to be approved than Black applicants with identical financial profiles

95%

White Approval Rate

White applicants with "low" credit scores of 640 were approved 95% of the time

<80%

Black Approval Rate

Black applicants with identical "low" credit scores were approved less than 80%

Four Main Types of AI Bias

1

Human Biases

Biases that originate from human perceptions, assumptions, and preferences

2

Data Bias

Biases introduced in the training data that alter representation of target populations

3

Algorithmic Bias

Biases inherent to data pre-processing or during model design and training

4

Model Deployment Biases

Biases that emerge when AI systems are implemented in real-world contexts

Human Biases

The dominant origin of biases observed in AI are human.
While rarely introduced deliberately, these reflect **historic or prevalent human perceptions, assumptions, or preferences** that can **manifest across various stages of AI model development**.

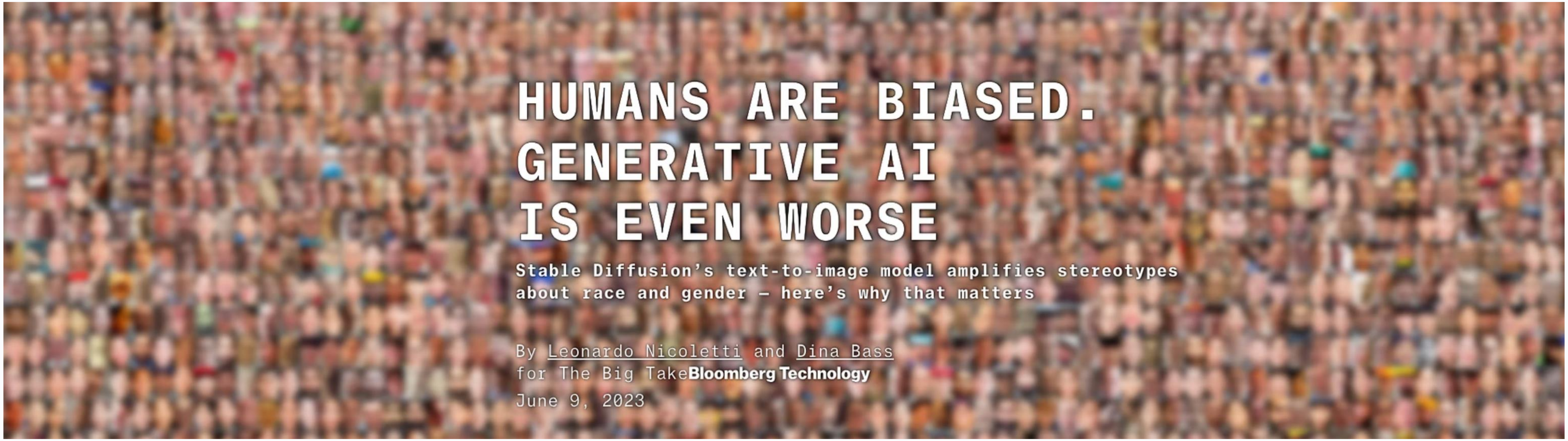
Implicit Bias

Implicit bias occurs **when subconscious attitudes or stereotypes about characteristics** like birth sex, gender identity, race, ethnicity, age, and socioeconomic status become embedded in how individuals behave or make decisions.

Human Bias Example

Bias in Generative AI

> *“We are essentially projecting a single worldview out into the world, instead of representing diverse kinds of cultures or visual identities”*



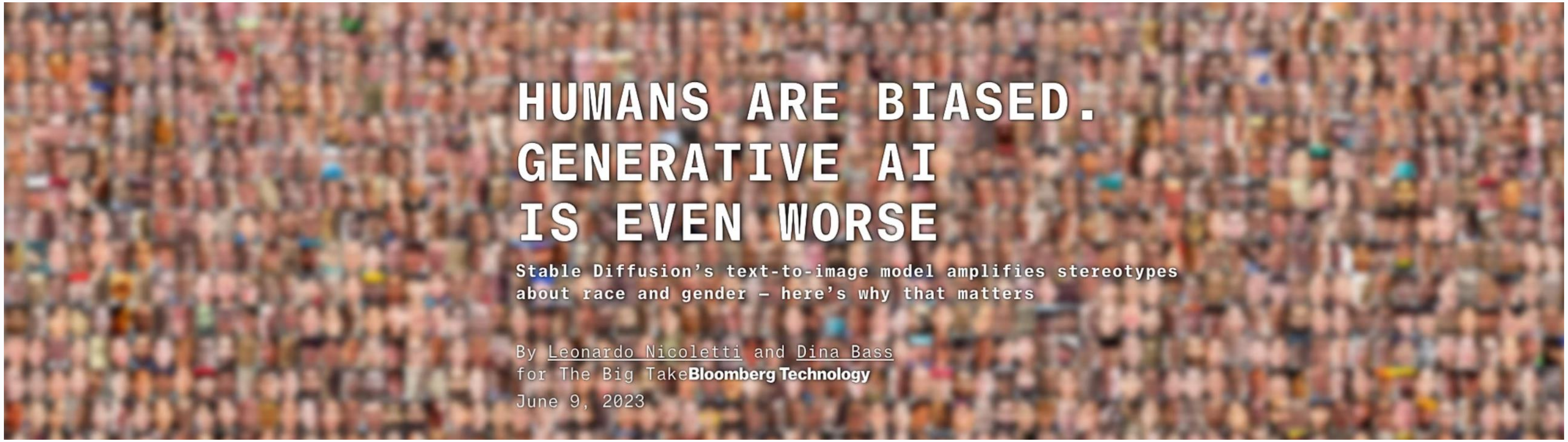
Human Bias Example

Bias in Generative AI

Stable Diffusion (a text-to-image model) was used to create representations of workers for 14 jobs:

- 7 jobs that are typically considered “high-paying” in the US
- 7 jobs that are considered “low-paying”

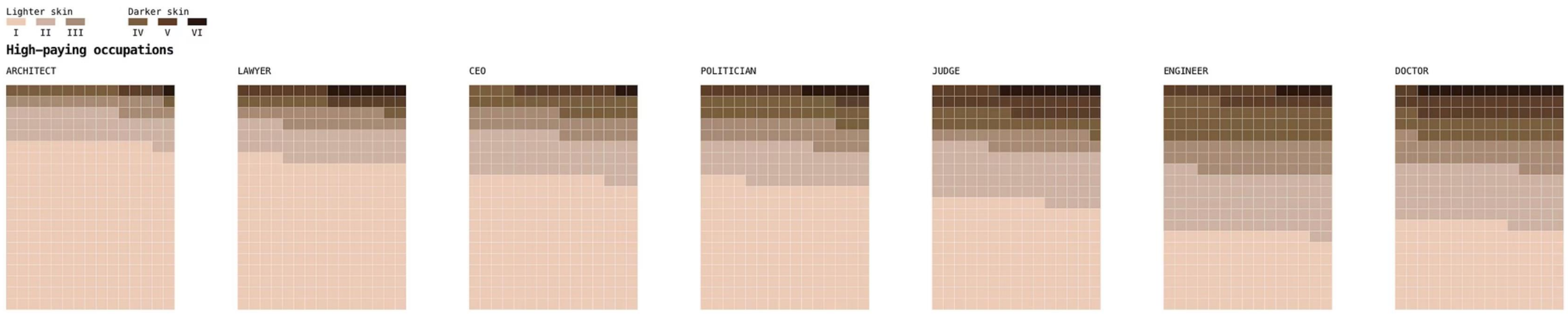
Plus three categories related to crime.



Human Bias Example

High-Paying Jobs

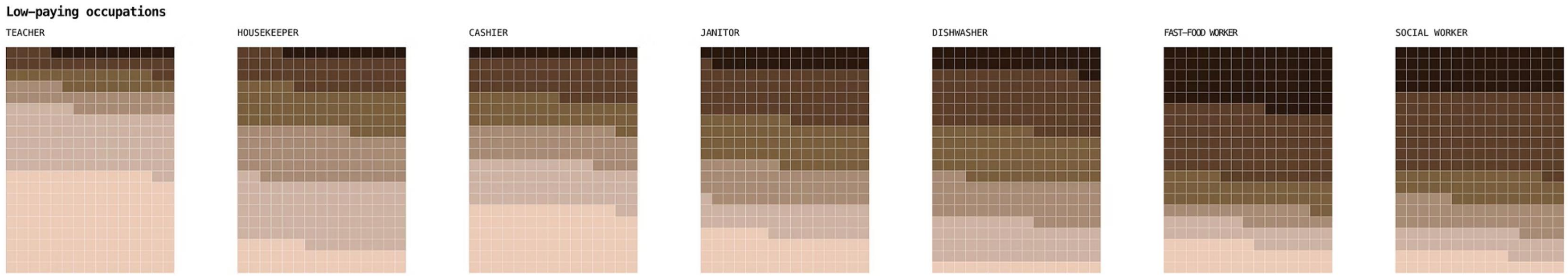
Image sets for high-paying jobs were dominated by subjects with **lighter skin tones**



Human Bias Example

Low-Paying Jobs

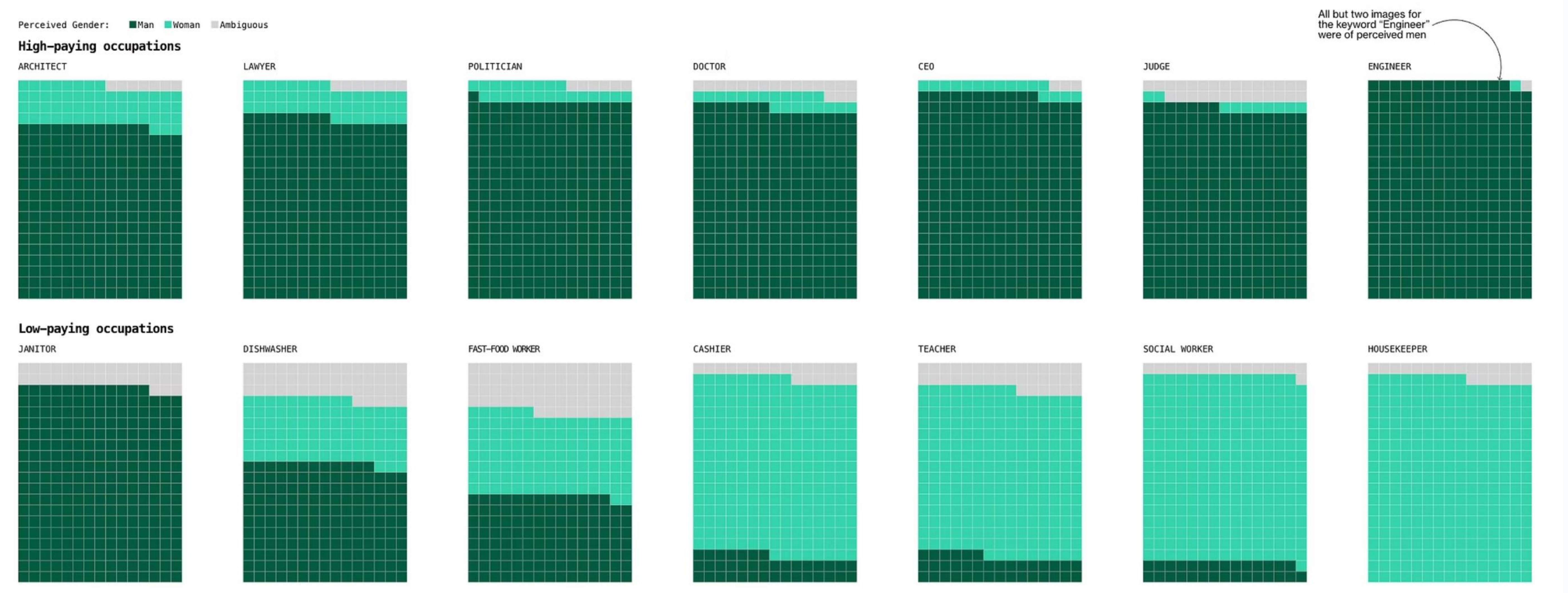
Subjects with **darker skin tones** were more commonly generated for "fast-food worker" and "social worker".



Human Bias Example

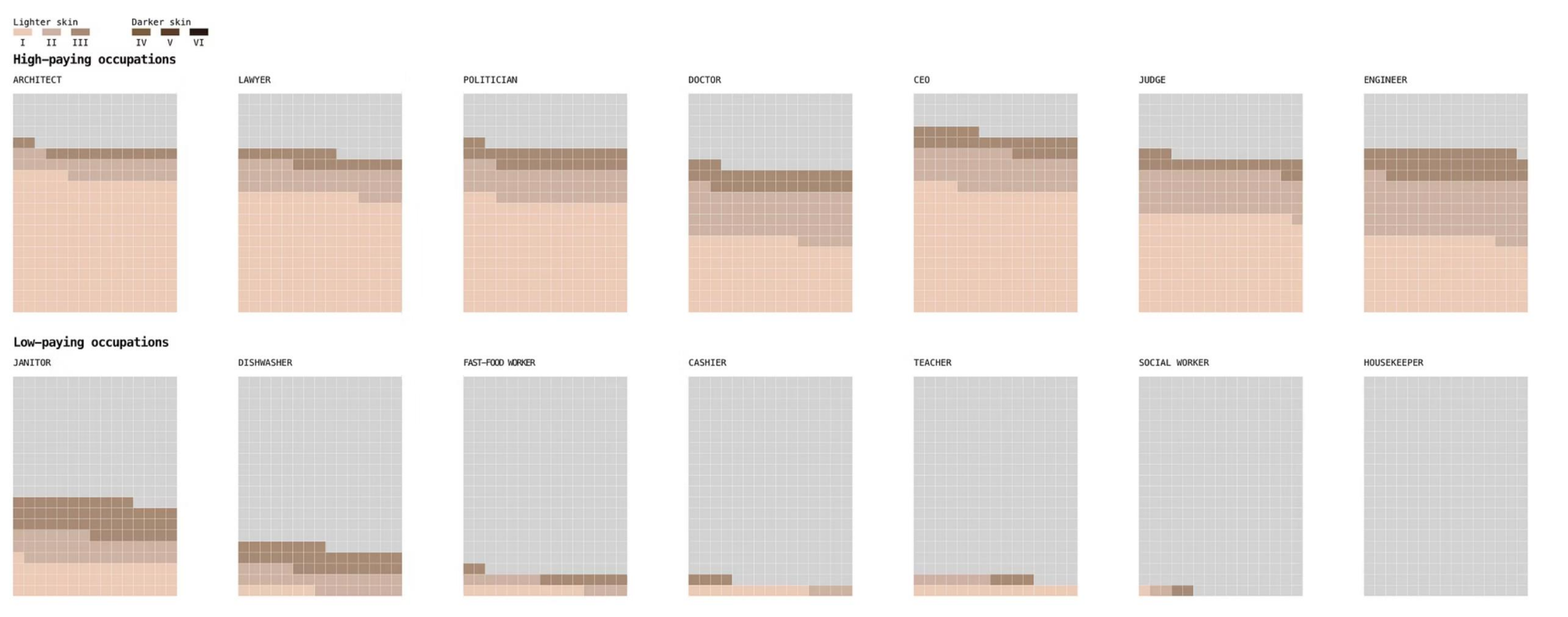
Gender Distribution

Most occupations were **dominated by men**, except for low-paying jobs like housekeeper and cashier



Human Bias Example

When considering both skin tone and gender, **men with lighter skin tones represented the majority** of subjects in every high-paying job, including "politician," "lawyer," "judge" and "CEO."



Data Bias

Beyond human influences, numerous additional **biases can be introduced to data** used in AI model training, **altering its representation of the target environment or population** and leading to skewed or unfair outcomes.

Representation Bias

Representation bias describes **a lack of sufficient diversity in training data** and is a dominant form of bias limiting the generalizability of AI models into unique environments or populations.

Data Bias Example

Discrimination in Hiring

AI-powered recruitment tools have been found to **favor male candidates over female applicants** due to biased training data.

Amazon's AI Hiring Tool: Amazon scrapped an AI recruitment system after it showed **bias against female candidates**, favoring resumes containing male-associated words, because **it was built on data from CVs submitted mostly by males**.

Algorithmic Bias

The biases are inherent to **the pre-processing of a training dataset** or during the **conceptual design**, **training, or validation phases** of an algorithm. They can stem from:

- incorrect or incomplete feature choices
- developer selection of algorithm processes
- poor validation on minority or rare subgroups

Algorithmic Bias

Aggregation Bias

An inappropriate combination of distinct groups or populations during data pre-processing leads to aggregation bias, where the **model's performance is only optimized for the majority**. In other words, it occurs when diverse population data is merged into a **single "one-size-fits-all" dataset**.

Imagine calculating "average student height" but:

- excluding special-needs students
- excluding international students
- filling missing heights with made-up values

The final average is meaningless — yet the model believes it's representative.

Algorithmic Bias

Feature selection bias

Feature selection bias is the **selective introduction or removal of variables** during model development **based on pre-conceived ideas or beliefs** surrounding the planned task.

Algorithmic Bias Example

Health Disparities

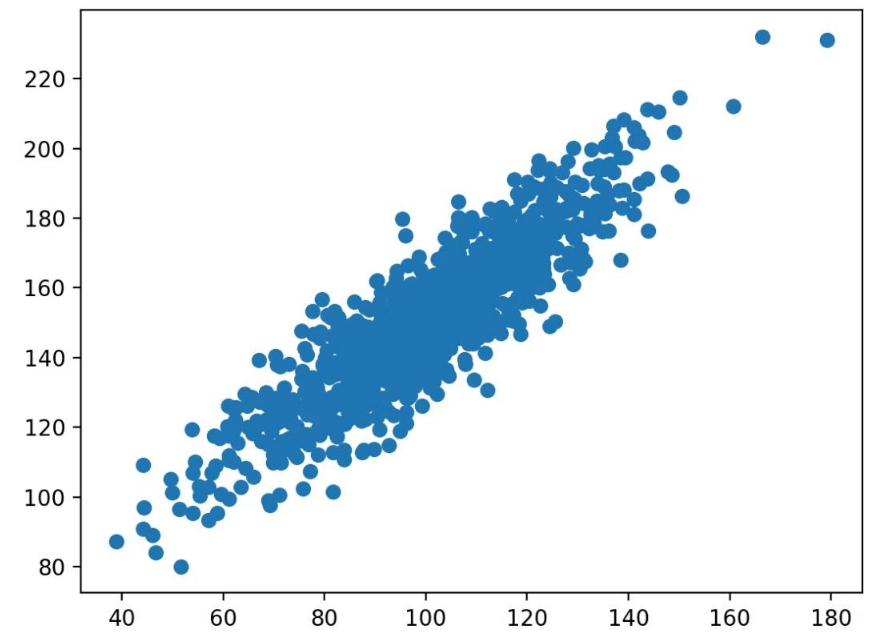
AI-based medical diagnostics have shown racial bias, leading to **misdiagnoses and incorrect treatment plans** for underrepresented populations.

> An AI used across several U.S. health systems exhibited bias by **prioritizing healthier white patients over sicker black patients** for additional care management **because it was trained on cost data, not care needs**.

Algorithmic Bias Challenge

If we delete the race and gender attributes from the data set, then it might seem that the system cannot discriminate on those attributes.

Unfortunately, machine learning models can predict latent variables (such as race and gender) given other correlated variables (such as zip code and occupation).



Model Deployment Biases

With AI system adoption incentives in the form of workflow efficiency, an **over-reliance** on AI systems and a **progressive de-skilling of the workforce** is of genuine concern.

Feedback Loop Bias

Feedback Loop bias occurs when **users consistently adopt and accept AI recommendations, even when inaccurate**, and these labels are then captured and reinforced by training cycles.

Biased prediction → biased human action → data reinforced → stronger bias

Model Deployment Biases Example

Feedback Loop Bias in Predictive Policing

If we utilize AI systems to predict "high-risk crime areas" to optimize resource allocation:

AI Prediction

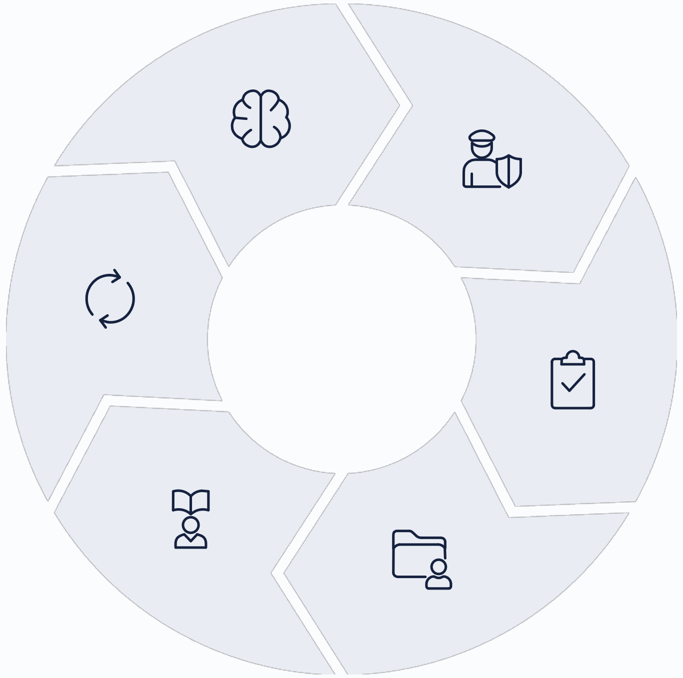
AI predicts a specific neighborhood is "high-risk" for crime based on historical data.

Self-Perpetuating Cycle

The model continues to predict high risk for the neighborhood, leading to even more patrols.

Model Learning

The AI model learns from this reinforced data, further solidifying its belief that the neighborhood has an even higher crime rate.



Increased Patrols

Police officers are dispatched to patrol the identified "high-risk" neighborhood more frequently.

More Incidents Recorded

With more officers present, a consequence is an increase in recorded incidents due to greater surveillance.

Data Reinforcement

These newly recorded "crime records" from the increased patrols are then fed back into the AI model's training data.

Strategies to Detect and Prevent AI Bias

Ensure Representative & Balanced Training Data

If your AI system is trained on historically biased financial data, it will likely replicate those biases unless proactive measures are taken to correct them.

Prevent Indirect Discrimination Through Feature Selection

Even if an AI model does not explicitly use race, gender, or other protected attributes, it can still learn to discriminate indirectly through features like ZIP codes, education levels.

Identify potential features that correlate strongly with protected attributes (e.g., using clustering or representation analysis to detect latent patterns between features)

Explainable AI

AI models should not operate as black boxes, especially for high-stakes financial decisions. Institutions must be able to explain why an AI system made specific decisions.

Strategies to Detect and Prevent AI Bias

Using Fairness KPIs

Use key performance indicators to verify equitable results across demographic groups:

- Demographic parity: Ensures positive outcomes occur at similar rates across different groups
- Equal opportunity: Ensures qualified individuals have the same likelihood of receiving positive outcomes
- Disparate impact analysis: Examines whether predictions result in significantly different outcomes for subgroups

Strengthening Human Oversight

Even with the best technology, AI cannot replace human judgment in financial decision-making.

Human oversight is essential to ensuring that AI operates fairly and ethically. Without human review, AI models can develop biases that go unchecked until they cause significant harm.

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Case Study:
Ethical dilemma of self-driving cars

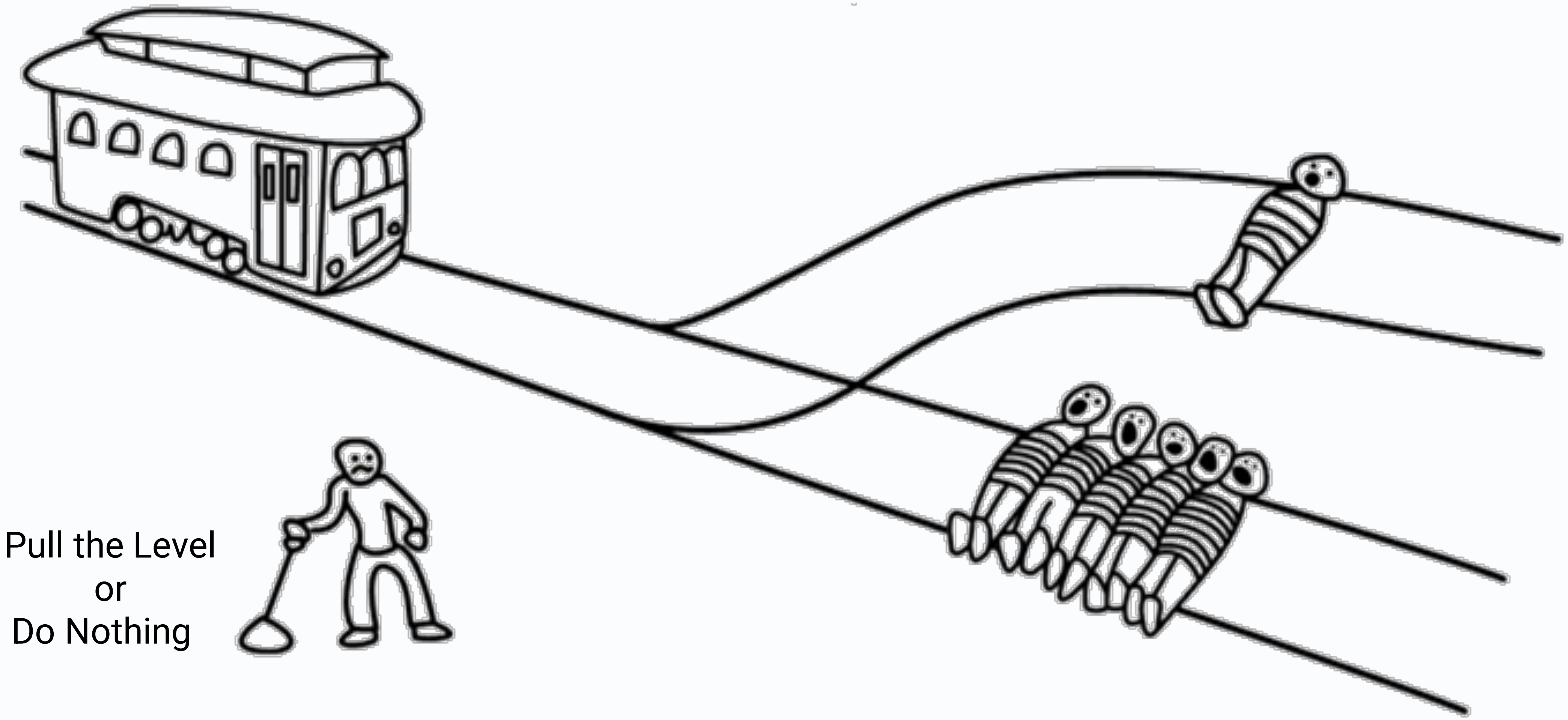
https://www.bilibili.com/video/BV1YeUmBLEDY/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

Case Study:
Absurd Trolley Problems

https://neal.fun/absurd-trolley-problems/

Scenario 1: Oh no! **A trolley is heading towards 5 people.**

You can pull the lever to divert it to the other track, killing 1 person instead.

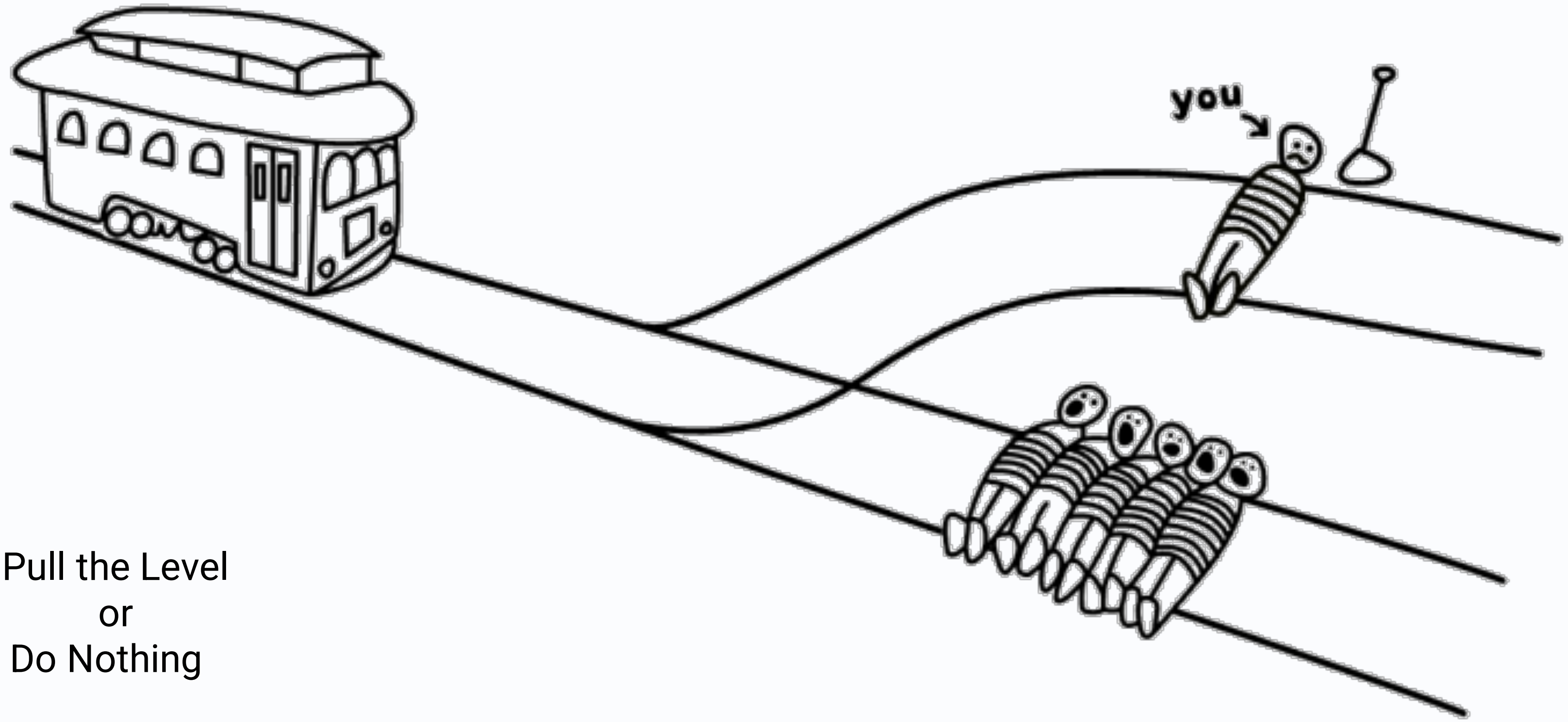


Q1: What do you do?

Q2: What will AI do?

Scenario 2: Oh no! **A trolley is heading towards 5 people.**

You can **pull the lever to divert it to the other track, sacrificing yourself instead**.

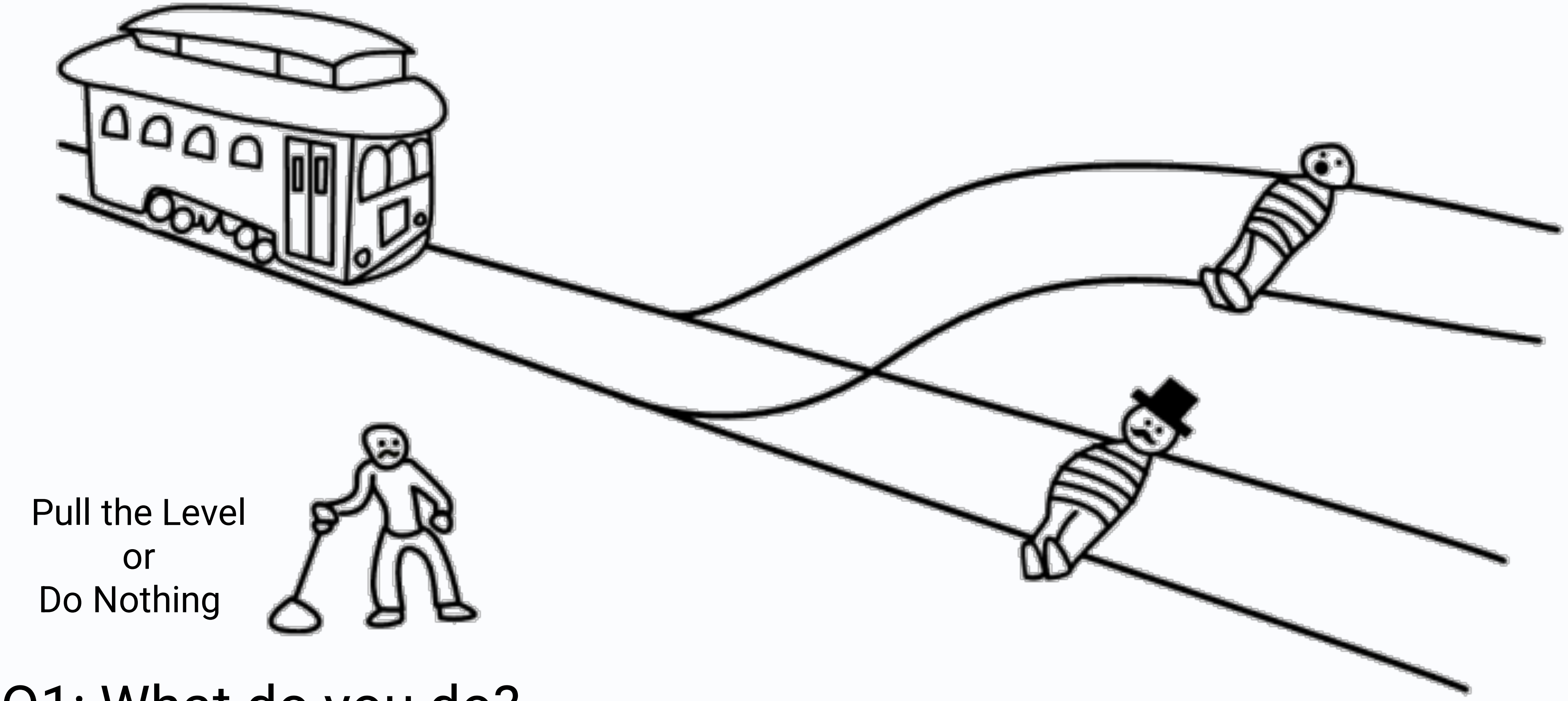


Pull the Level
or
Do Nothing

Q1: What do you do?

Q2: What will AI do?

Scenario 3: Oh no! **A trolley is heading towards a rich man. The rich man offers you $500,000 to pull the lever**, which would divert the trolley **and kill someone else**.

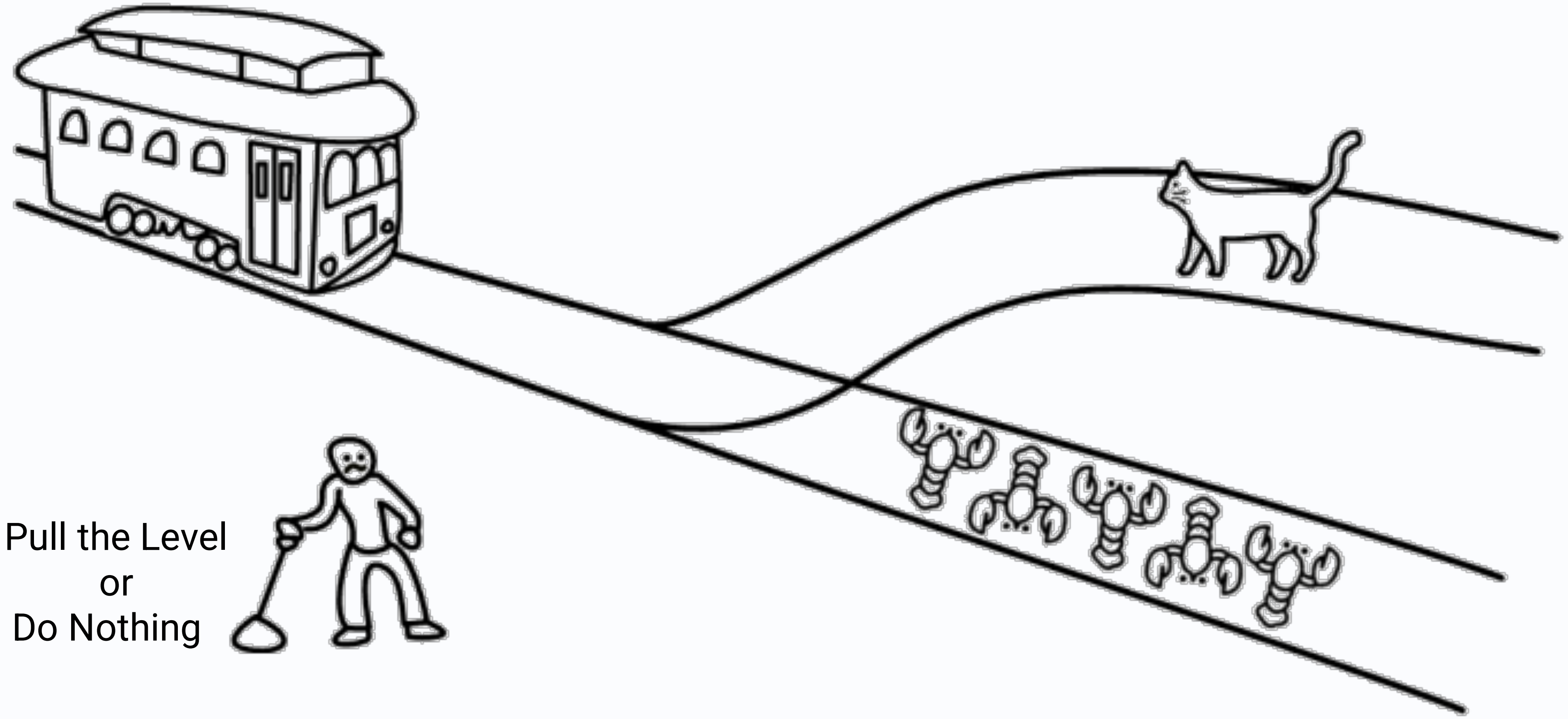


Q1: What do you do?

Q2: What will AI do?

Scenario 4: Oh no! A trolley is heading towards **5 lobsters.**

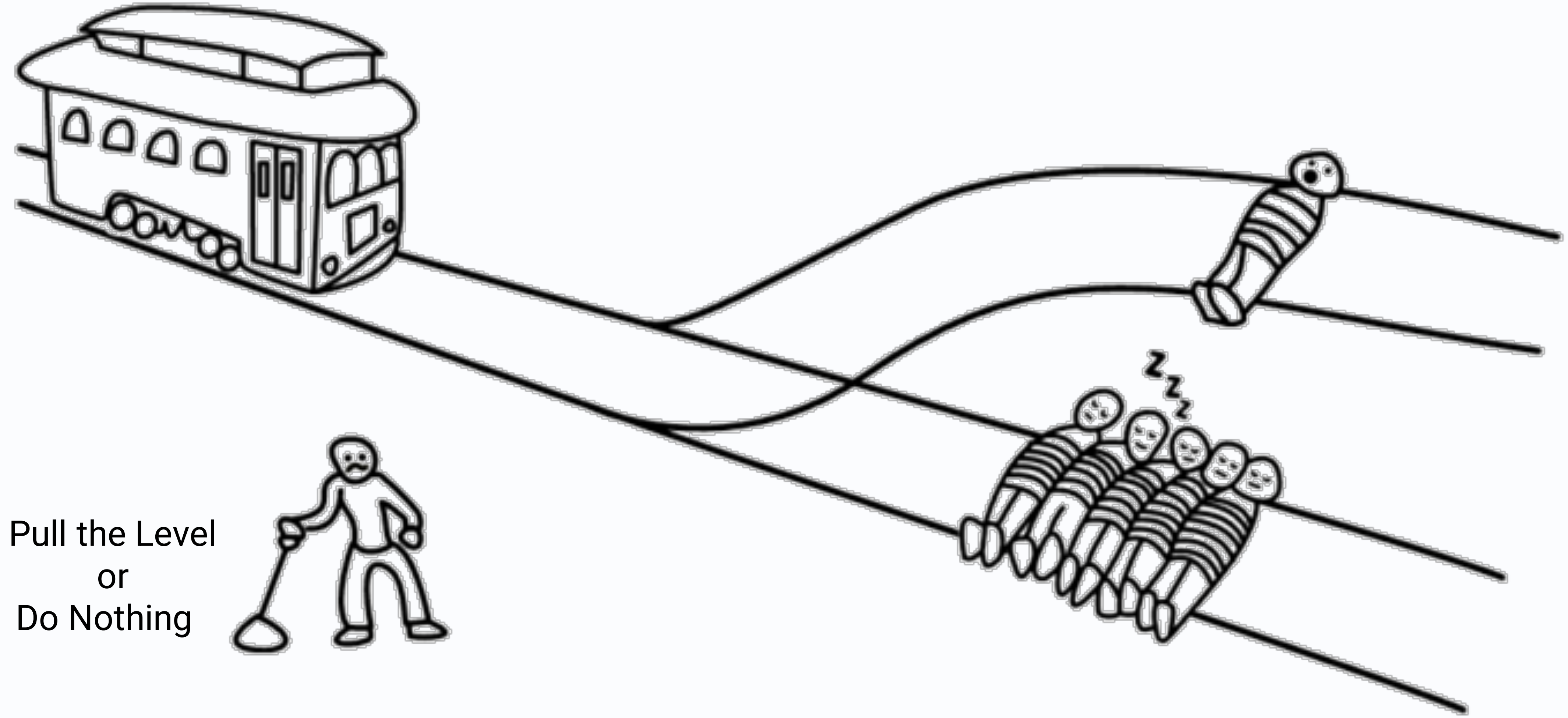
You can **pull the lever to divert it to the other track, running over a cat** instead.



Q1: What do you do?

Q2: What will AI do?

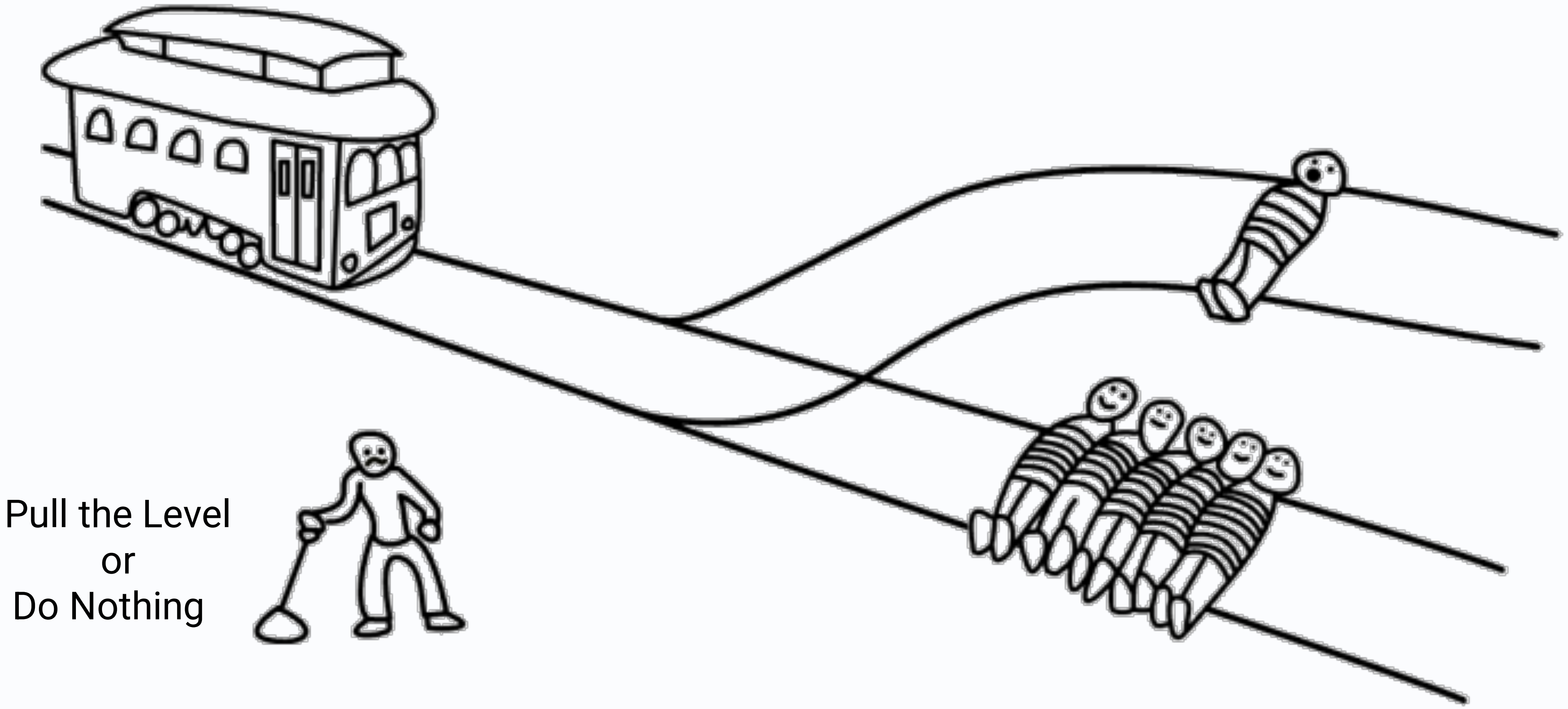
Scenario 5: Oh no! A trolley is **heading towards 5 people who are sleeping and won't feel pain**. You can **pull the lever to divert it to the other track, running over someone who is wide awake** instead.



Q1: What do you do?

Q2: What will AI do?

Scenario 6: Oh no! A trolley is heading towards **5 people who tied themselves to the track**. You can **pull the lever to divert it to the other track, killing 1 person who accidentally tripped onto the track** instead.



Q1: What do you do?

Q2: What will AI do?

Case Study:
AI Responses to Trolley Problems

https://www.bilibili.com/video/BV1eeUmBLEWD/?vd_source=b0e15bdbda6a17abcc8c653915e36a8f

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https://www.moralmachine.net/y

Questions?

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